

# CHAPTER 1

## INTRODUCTION

### 1.1 PURPOSE OF THIS MANUAL

This manual was prepared to assist entry-level engineers in the design of air-conditioning systems. It is also usable—in conjunction with fundamental HVAC&R resource materials—as a senior- or graduate-level text for a university course in HVAC system design. This manual was intended to fill the void between theory and practice, to bridge the gap between real-world design practices and the theoretical knowledge acquired in the typical college course or textbook. Courses and texts usually concentrate on theoretical calculations and analytical procedures or they focus upon the design of components. This manual focuses upon applications.

The manual has two main parts: (1) a narrative description of design procedures and criteria organized into ten chapters and (2) six appendices with illustrative examples presented in greater detail.

The user/reader should be familiar with the general concepts of HVAC&R equipment and possess or have access to the four-volume *ASHRAE Handbook* series and appropriate ASHRAE special publications to obtain grounding in the fundamentals of HVAC&R system design. Information contained in the *Handbooks* and in special publications is referenced—but not generally repeated—herein. In addition to specific references cited throughout the manual, a list of general references (essentially a bibliography) is presented at the end of this chapter.

The most difficult task in any design problem is how to begin. The entry-level professional does not have experience from similar projects to fall back on and is frequently at a loss as to where to start a design. To assist the reader in this task, a step-by-step sequence of design procedures is outlined for a number of systems.

be benchmarked with a criterion that requires compliance with *ANSI/ASHRAE/IESNA Standard 90.1, Energy Standard for Buildings Except Low-Rise Residential Buildings*.

*Design validation* involves the use of a wide range of estimates, calculations, simulations, and related techniques to confirm that a chosen design option will in fact meet the appropriate design criteria. Design validation is essential to successful design; otherwise there is no connection between design intent and design decisions. *Pre- and post-occupancy validations* are also important to ensure that the construction process and ensuing operational procedures have delivered design intent. Such validations are a key aspect of building commissioning.

## 2.2 DESIGN VERSUS ANALYSIS

Anyone who has taken a course in mathematics or any of the physical sciences is familiar with the process of analysis. In a typical analysis, a set of parameters is given that completely describes a problem, and the solution (even if difficult to obtain) is unique. There is only one *correct* solution to the problem; all other answers are *wrong*.

Design problems are inherently different—much different. A design problem may or may not be completely defined (some of the parameters may be missing) and there are any number of potentially acceptable answers. Some solutions may be better than others, but there is no such thing as a single right answer to a design problem. There are degrees of quality to design problem solutions. Some solutions may be better (often in a qualitative or conceptual sense) than others from a particular viewpoint. For a different context or client, other solutions may be better. It is important to clearly understand the difference between *analysis* and *design*. If you are used to looking for *the* correct answer to a problem (via analysis), and are suddenly faced with problems that have several acceptable answers (via design), how do you decide which solution to select? Learn to use your judgment (or the advice of experienced colleagues) to weigh the merits of a number of solutions that seem to work for a particular design problem in order to select the best among them.

Figures 2-1 and 2-2 illustrate the analysis and design processes, respectively. Analysis proceeds in a generally unidirectional flow from given data to final answer with the aid of certain analytical tools. Design, however, is an iterative process. Although there are certain “givens” to start with, they are often not immutable but

A heavy envelope of brick or masonry provides thermal mass; such mass, by shifting loads, affects the operating cycle of the heating and cooling equipment. The greater the internal building mass, the less extensive are the indoor temperature excursions. The placement of thermal insulation, whether on the exterior or the interior side of the envelope, affects heating and cooling load patterns. The envelope construction and the location of the vapor retarder (if required) also determine the transfer of moisture between the building spaces and the outdoors, thus affecting humidification and dehumidification requirements. The interaction of HVAC system and envelope relative to moisture flows is especially critical to building success in hot-humid and cold climates.

A building can be configured to provide solar shading and wind protection, which will influence heating and cooling loads and control strategies. By reducing transmission and radiation transfer at the perimeter, the envelope reduces the influence of climate on the building interior and thereby affects the designation of thermal zones. If loads in a building do not vary much with time of day, solar intensity, and wind velocity and direction, an HVAC system can be less complex and less costly.

Another effect of the envelope on HVAC&R design is the use of daylighting to offset electric lighting. Less heat is gained with well-designed daylighting than with electric lighting for the same illumination levels, so cooling requirements are reduced.

**2.4.1.2 How the HVAC&R System Affects a Building Envelope.** The HVAC&R system can affect the appearance of the building envelope. Outdoor air intake louvers in walls, window air conditioners, and through-the-wall units obviously affect the appearance of a building. This may limit system selection options where the resulting appearance is not acceptable to the architect or client.

Cooling towers, usually located on the roof, affect the physical appearance of a building. Other rooftop equipment, such as packaged air conditioners, air-cooled condensers, and air intake and exhaust hoods/fans, may have a similar effect. Coordination of intake/exhaust locations and louvers will be required for ventilation heat recovery systems. Attractive architectural enclosures can reduce the aesthetic disadvantage of much rooftop equipment. Under certain conditions, such equipment can be placed at ground level and camouflaged by landscaping. Locating condenser/compressor units for buildings with multiple split systems can be espe-

- Ensure that there is ready access for routine maintenance tasks, such as changing air filters.
- Specify accessible service shutoff drain and vent valves for all water coils and risers.
- Incorporate appropriate chemical treatment equipment for open- and closed-loop water systems.

### 2.5.3 Noise and Vibration Control

Noise and vibration from HVAC&R equipment can be reduced by proper selection of equipment, vibration control, and the interposition of sound attenuators and barriers. Residual sound may still be objectionable, however, when equipment is located near occupied areas. HVAC&R and architectural design decisions must go hand-in-hand since acoustical control may involve equipment location, floor and wall assemblies, and room function. HVAC&R equipment locations may influence spatial planning—so that areas requiring a low-noise environment are not located next to major or noisy equipment. The HVAC engineer should alert other members of the building design team regarding the location of noisy equipment. On large, acoustically sensitive buildings (e.g., theaters, museums), an acoustical consultant should be part of the design team.

Noise and vibration transmission to an occupied space by system components will be an important consideration in system selection and design. Even after a system has been selected, component selection will significantly affect system acoustical performance. Noise can be transmitted to occupied spaces from central station equipment along several airborne paths, through air or water flows, along the walls of ducts or pipes, or through the building structure. If central station pumps or fans are used, each of these paths must be analyzed and the transmission of sound and vibration reduced to an acceptable level. Supply air outlets (and other terminal devices) must be selected to provide appropriate acoustical performance.

An initial step in noise control is to establish noise criteria for all spaces. These criteria should be communicated to the client early in the design process. All noise-generating sources within the air-conditioning system must be identified. Then, the effects of those sources can be controlled by

- equipment selection,
- air and water distribution system design,
- structural/architectural containment,
- dissipative absorption along the sound path, and
- isolation from the occupied space, where feasible.

[950 L/s]) and those with higher pressure differentials produce greater noise levels. Pressure-reducing devices should be installed in the duct system with sufficient downstream ductwork to absorb the sound generated by the device. Large terminal units with pressure-reducing devices should not be installed in occupied spaces without considering acoustical treatment downstream and in the radiated sound path from the terminal to the room.

Sound can travel through ductwork from one room to another. For example, an air-conditioning system that serves a series of music practice rooms will require ductwork with sound baffles between rooms, lined ducts, or ample duct turns to attenuate noise. Noise control will influence duct configuration, size, and system static pressure.

The sound produced by room terminal equipment cannot be easily reduced. Control of this potential problem starts with system selection and entails careful equipment selection and sizing to achieve the noise criteria for a given conditioned space. The more moving parts in a terminal, the noisier it will be. Air-cooled unitary terminal equipment is likely to be near the high end of the noise scale. Water-cooled terminals, including water-source unitary terminals, can be significantly quieter. Air terminal equipment, in ascending order of noisiness, include air diffusers, variable-air-volume boxes, fan-coil units, high-induction-ratio terminals, and packaged terminal air conditioners. Continuous terminal noise is usually less annoying than intermittent or alternating noise. Thus, ON/OFF control of terminal refrigeration equipment and air-circulating fans may produce annoyance even though the sound pressure level during equipment operation is within acceptable limits.

Terminal equipment, because of its location, provides the fewest options for acoustical mitigation. The solution is essentially in the selection of the equipment itself. Greater opportunities for noise control through attenuation (e.g., duct lining) and barriers (e.g., solid ceilings) are possible when terminal equipment can be located outside the occupied space (for example, placing air-mixing boxes above a ceiling with low-velocity ducts connecting to diffusers). Air ducts passing through adjacent rooms can be transmission channels for cross-talk, as can unsealed openings around ducts or pipes. Cross-talk through such paths can be controlled through building design. Occasionally, partitioning will be located so as to divide a room terminal or outlet. This creates a virtually uncontrollable path for sound transmission between rooms.

Design of piping from the tower sump to the pump should be undertaken with caution. Place the sump level above the top of the pump casing to provide positive prime. Pitch all piping up, either to the tower or to the pump suction, to eliminate air pockets. Suction strainers should be equipped with inlet and outlet gauges that indicate when cleaning is required. Normally, piping is sized to yield water velocities between 5 and 12 fps [1.5 and 3.7 m/s]. Use friction factors for aged or old pipes.

In cooling tower installations, ascertain that sufficient NPSH (net positive suction head—see Section 5.5.2) for the condenser-water pump is provided, and design the connections so that the flow through each cooling tower has the same pressure drop. Install equalizing lines of proper size between towers; otherwise, one tower will fill and the other one will overflow. Winterize the tower piping if necessary or provide for drain-down when pumps shut off. In the latter case, the isolation valves must be located in a heated space. Provide valving or sump pump isolation so that any single cell can be serviced while the others are active.

#### 5.4.6 Evaporative Condensers

Evaporative condensers are used to reject heat to the outdoor air by means of the evaporation of water. Water is sprayed on the condenser coils, and the resultant heat of vaporization, combined with any sensible heat effect, cools the liquid inside the coils. The residual spray water is usually recirculated. An evaporative condenser uses much less makeup water than a cooling tower in a conventional water-cooled condenser arrangement (approximately 1.6 to 2.0 gph/ton [0.5 to 0.6 mL/s per kW] by evaporation plus 0.5 to 0.6 gph/ton [0.15 to 0.18 mL/s per kW] for bleed water). Evaporative condensers are used primarily as refrigeration system condensers. In such applications, they should be located as close as possible to the refrigeration machinery to keep the refrigerant lines short. In the winter, the water can often be shut off with the dry coil providing adequate heat rejection. Evaporative condensers are also used to cool recirculating water in closed-loop water-source heat pump systems (see Chapter 7).

Figure 5-8 shows two evaporative condenser types using city, well, or river makeup water. Piping should be sized in accordance with the principles outlined in Section 5.4.5 with flow velocities of 5 to 10 fps [1.5 to 3.0 m/s]. Where city water is used, a pump is usually not required since the water arrives under pressure. For well or river water, pumps may be necessary, in which case the